

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	PNT2022TMID124356
Project Title	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution uses Linear Regression to predict HDI scores using socio-economic indicators.

Project Overview	
Objective	The proposed solution uses Linear Regression to predict HDI scores using socio-economic indicators.
Scope	Accurate HDI predictions with category classification
Problem Statement	
Description	Data Collection Data Cleaning Feature Engineering Model Training Model Evaluation Web Deployment

Impact	Solving these issues will result in improved operational efficiency, reduced risks, and an overall enhancement in the lending process, contributing to customer satisfaction and organizational success.
Proposed Solution	
Approach	Employing machine learning techniques to analyze and predict creditworthiness, creating a dynamic and adaptable loan approval system.
Key Features	- Implementation of a machine learning-based credit assessment model.



	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		

Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv